

Mahima Agarwal

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Education

Cluster Innovation Centre, University of Delhi

Bachelor of Technology in Information Technology and Mathematical Innovations

Expected June 2026

New Delhi

Open Source Experience

Wikimedia Foundation

Outreachy Intern

May 2024 – Aug 2024

Remote

- Developed a **data visualization tool** for tracking the quality evolution of Wikipedia articles maintained by different WikiProjects.
- Leveraged **Wikimedia APIs**, **Wiki Quarry**, and **Toolforge** with **SSH access** to manage and optimize data workflows in a remote server environment.
- Handled millions of data rows using advanced **query optimization** and **data warehousing** techniques to ensure scalable analysis.
- Integrated **language-agnostic model predictions** into the tool to provide more accurate and universal quality assessments.
- Conducted extensive pre-development **research** to align with Wikimedia standards, presenting outcomes at the **IIIT Hyderabad Hackathon** and the **Wikimedia Technology Summit 2024**.

Key Learnings: Statistical Graphics, Data Analysis, Data Warehousing, Query Optimization, PAWS, SSH, MariaDB, API Integration, Toolforge, Pre-project Research, Documentation, Presentation Skills.

Research Experience

Research Intern

Using Reservoir Computing (RC) for Image Encryption and Forecasting of a Hyperchaotic Financial Model

Delhi

- Under review at Springer – Multimedia Tools and Applications.**
- Developed an advanced **image encryption** scheme using **RC models** to enhance security and efficiency in transmitting visual data.
- Integrated **chaos theory** and **fractional-order systems** to improve encryption for real-time applications.
- Applied **RC-generated chaotic sequences** for diffusion, ensuring high unpredictability throughout encryption.
- Utilized a **two-key system**: the first key generates system parameters (**rho**, **phi**), while the second controls chaotic diffusion for enhanced randomness.
- Implemented **pixel shuffling** and **bitwise XOR operations** to scramble pixel values, followed by reshaping into the original image format.
- Validated encryption robustness against **differential attacks** using **NPCR**, **UACI**, and **histogram variance metrics**.
- Key Learnings:** Advanced Cryptography, Reservoir Computing, Chaos Theory, Fractional-Order Systems, Security Metrics, NPCR, UACI, Real-Time Transmission.

C4GT Intern

IIIT Delhi – Hand Gesture Occlusion Research Project

Jun 2025 – Aug 2025

Delhi

- Built a scalable pipeline to extract **3D hand landmarks** from RGB images and compare gestures using **point cloud similarity metrics**.
- Used **MediaPipe** for landmark detection, **Open3D** for point cloud generation/export, and **PyTorch3D** for computing **Chamfer Distance**.
- Enabled **parallel GPU-accelerated processing** for efficient computation on large gesture datasets and exported results to CSV.
- Visualized 3D hand landmarks and analyzed gesture similarity patterns using histograms and qualitative inspection.
- Key Learnings:** Computer Vision, 3D Point Clouds, Gesture Recognition, Chamfer Distance, PyTorch3D, MediaPipe, Open3D, Parallelization, Data Visualization.

CyPSi Laboratory

Research Intern

Feb 2024 – Apr 2024

Delhi

- Developed a **retinal image segmentation system** using advanced **deep learning** techniques to support medical diagnostics.

- Implemented **U-Net** and **Convolutional Neural Networks (CNNs)** for high-precision segmentation of fundus images.
- Applied **GAN-based data augmentation** to improve model generalization and robustness.
- **Key Learnings:** Deep Learning for Medical Imaging, CNNs, U-Net, GANs, Python (TensorFlow/PyTorch), Medical Dataset Handling.

Projects

Semantic Search Engine for NIC | *Python, FAISS, Hugging Face Transformers, Sentence-BERT, Streamlit*

- Developed a **semantic search engine** to efficiently search across NIC's policy documents and internal data.
- Implemented **Sentence-BERT** embeddings and integrated them with **FAISS** for high-performance similarity search.
- Built an intuitive **Streamlit-based UI** to provide real-time, context-aware search results.
- Optimized embedding storage and retrieval to ensure low-latency search over large datasets.
- **Key Learnings:** Semantic Search, FAISS Indexing, Sentence-BERT, Hugging Face Transformers, Streamlit App Development, Information Retrieval.

PPML using Paillier Cryptosystem | *NumPy, Python, Matplotlib, pandas, Scikit-Learn*

- Showcased the effectiveness of the **PPML** approach in balancing accurate predictions and privacy preservation.
- Integrated the **Paillier cryptosystem** and **linear regression** to derive meaningful insights without compromising individuals' privacy.
- **Key Learnings:** Privacy-Preserving Machine Learning (PPML), Homomorphic Encryption, Linear Regression, Data Privacy, Python Libraries (NumPy, pandas, Scikit-Learn).

Chaos-Lock | *MATLAB, Python*

- Developed an **encryption model** in MATLAB based on the application of the **Logistic Map Equation** and its chaotic properties.
- Implemented a **PRNG** (Pseudo-Random Number Generator) based on the chaotic dynamics of the Logistic Map Equation.
- **Key Learnings:** Chaotic Systems, Logistic Map, PRNG Development, MATLAB Programming, Advanced Encryption Techniques.

Skills

Programming Languages: Python, C, C++, Java, JavaScript, MATLAB, HTML5

Libraries & Frameworks: NumPy, Pandas, Scikit-Learn, TensorFlow, PyTorch, OpenCV, FAISS, Hugging Face Transformers, Sentence-BERT, Streamlit, Tkinter

Development Tools & Platforms: Node.js, MongoDB, Docker, Git, GitHub, Toolforge, PAWS

Technologies & Concepts: Machine Learning, Deep Learning, Semantic Search, Information Retrieval, GANs, Cryptography, Reservoir Computing, Data Warehousing

Soft Skills: Research, Problem-Solving, Time Management, Cross-Team Collaboration, Presentation Skills